

SENSATIONAL!

Disrupter: Sensors (Automated Systems, Self Driving Cars, IoT)

Disrupted: Everything

What Year: 2020 to 2025

When it comes to disruptors, we're starting off with the one we think will kill many businesses, and create a whole new way of living. Sensors, those little tiny things that are being printed out now, they're that small, are going to make the world so different, that if you went to sleep today and woke up in 2030, you'd think you slept for a century because of how different life will be.

There are many fields of science and technology that sensors will disrupt, most notably, the auto industry, agriculture, manufacturing, and medicine. All of this will be enabled by what everyone calls the IoT (Internet of Things, though we sometimes call it the IoE – Internet of Everything – but more on that later). Sensors are going to go everywhere. Into the food we grow, the things we build, and yes, even us. Let's look at the three biggest disruptions we believe sensors will cause.

THE AUTO INDUSTRY

This one's expected. It's been too long since the last real shake up of the industry. With the oil industry already under threat from electric cars, the auto industry itself is under threat, and from Google, of all companies. Let us try to illustrate how.

Google is currently working on driverless cars. Many others are, but Google is way ahead of everyone else in this particular field. Sensors are the mainstay of getting a car to go driverless. Since a computer can only sense through sensors, Google's using all the sensors it can to get the computer to drive safely, yet not too slowly.

The Mars Rovers, for instance, are an example of driverless vehicles, because they had to be. They would be on an alien world, too far away to be driven remotely, no spares or people to change your tyres or do repairs... That's why they crawl around Mars at excruciatingly slow speeds, more careful of every inch they move than the most paranoid human that

ever lived. That's OK for NASA though, they're not trying to get from point A to point B on Mars, and aren't late for any appointments with the Martians.

Although you'd think Google's job would be far easier, given that its cars drive on nice smooth roads and get regularly serviced, the fact is what they've done is way more impressive, because they already have cars driving about in almost everyday traffic, amidst human drivers who don't always follow the rules, and doing so at speeds of a normal commute.

When the technology is perfected, driverless cars will have to drive alongside humans for many more years to come, and they will need to do so safely.



A prototype of a self driving car by Google

Now assume that day has arrived, and you now own a spanking new driverless car. After all the initial excitement has died down, and they're as common on a road as regular taxis, you're going to be bored, and get back to doing what you do best, which is fiddle with your phone or computing device, and get back to work or play. How do you normally use your car? Apart from the very, very occasional pleasure drive, living in a city probably means you use vehicles to get you from point A to B. If that vehicle is your private car, it's going to be parked somewhere, while you do whatever it is you went to

point B to do – usually work, or school, or whatever. Then when you're done, you either hop into your car and drive back to point A (home), or at best go to point C to buy some stuff, and sometimes point D to catch up with friends, before heading home to point A again.

Seems like you'd still need a car, even after you no longer need to drive it, right? Perhaps. Now consider your parents, or family at home. They also need to commute. They might have a second car, if you can afford it. With a driverless car, it could drop you to point B, go back to point A and take the family around as well. It would certainly reduce the amount of cars on the road. Maybe not all, but certainly some homes would make do with just one car.

Now consider that taxi companies could drop prices drastically, because their biggest cost at the moment is the human drivers. If all they were paying for was fuel and maintenance, they would slash prices to get you to use their cars. They would also be able to afford more cars, which would make them cover more and more areas easily and make availability a breeze. You might choose, like many will, to use only the cheap taxi service. Heck, you might choose to use a cheaper and newer sharing taxi service, which can now accommodate four people in a car (two in the back two in front as there's no driver) with great comfort and luxury seats. All that comfort by just compromising on a few minutes as the taxi takes a few short detours to pick up people headed in approximately the same direction as you.

It's not unthinkable to imagine that many people would choose to do without the headache of owning and maintaining their own cars, when day or night they can get a car to take them where they want to go on demand. What this translates to is a huge reduction of the total amount of cars. We don't foresee a great reduction of the amount of cars on the road. What will